Initials:_____

BME 105: Genetics in the Genomics Era
Winter 2018: Midterm 2

Name: _____

Student ID:_____

Before starting, write your initials on the top of each page.

You can use the back-side of the pages for scratch, but we will not grade answers written on the back-side of the page.

Section	**Points**
Part I: One- or two-word answers and True or False	/22
Part II: Multiple choice	/24
Part III: Short answer	/24
Part IV: Extended calculation	/30
Total Points	**/100**

Second Letter

1st letter	U		C		A		G		3rd letter
U	UUU UUC UUA UUG	Phe Leu	UCU UCC UCA UCG	Ser	UAU UAC UAA UAG	Tyr Stop Stop	UGU UGC UGA UGG	Cys Stop Trp	U C A G
C	CUU CUC CUA CUG	Leu	CCU CCC CCA CCG	Pro	CAU CAC CAA CAG	His Gln	CGU CGC CGA CGG	Arg	U C A G
A	AUU AUC AUA AUG	Ile Met	ACU ACC ACA ACG	Thr	AAU AAC AAA AAG	Asn Lys	AGU AGC AGA AGG	Ser Arg	U C A G
G	GUU GUC GUA GUG	Val	GCU GCC GCA GCG	Ala	GAU GAC GAA GAG	Asp Glu	GGU GGC GGA GGG	Gly	U C A G

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Part I: One- or two-word answers and True and False

In the space provided next to each definition, clearly write the appropriate term.

1. _____one gene affecting more than one phenotype
2. _____ pattern of expression depends on which parent transmitted the allele
3. _____ mutation that does not change the identity of the amino acid
4. _____ site of DNA methylation
5. _____ the alleles of one gene masks the effects of the alleles of another gene
6. _____ a gain-of-function allele that produces a new function
7. _____ another mutation that alleviates a phenotypic abnormality

8. Angelman syndrome is caused by a mutation in the maternal copy of an autosomal paternally imprinted gene. Label the following statement as **true** or **false**; assuming that the trait is 100% penetrant.

    a. half of the sons of affected males will show the syndrome.

    b. half of the sons of affected females will show the syndrome.

    c. half of the daughters of affected males will show the syndrome.

    d. half of the daughters of affected females will show the syndrome.

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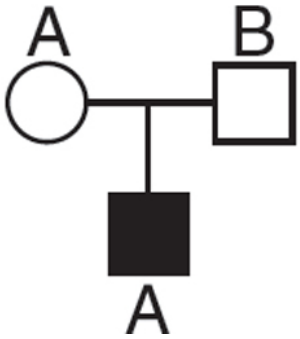
Part II: Multiple choice (circle your answer. If there is more than one correct answer, circle all correct answers)

9. You are interested in identifying all promoters in the A549 cell line (a lung cancer cell line). Which experimental approach would you most likely use?

    a. ATAC-Seq
    b. ChIP-Seq
    c. Bisulfite sequencing

10. Consider the following pedigree from a human family containing a male with Klinefelter syndrome (a set of abnormalities seen in XXY individuals; indicated with a shaded box). A and B refer to alleles of the X-linked *G6PD* gene. There are only two phenotypes associated with the *G6PD* gene, A or B. The phenotypes of each individual are shown on the pedigree. Indicate if nondisjunction occurred in the mother or father of the son with Klinefelter syndrome.



    a. mother
    b. father
    c. cannot be determined

11. A woman with type A blood has a child with type A blood. Which of the following men could not be the father of this child?
    a. A
    b. B
    c. AB
    d. All could be the father

12. Two wild-type fragments of human genomic DNA from the long arms of two nonhomologous acrocentric chromosomes are shown below. On both fragments the centromere-to-telomere direction is left-to-right. The red lines numbered 1,2 and 3 indicate the positions of breakpoints that could be caused ionizing radiation. Which two breakpoints could yield a reciprocal translocation?

```
5'  TCGATTCCGGAAAGCTTAGTTTCCCGGGACGTATTGCCAACCTAGGTAAGCGCCGAATATCCATGGGCACC  3'
3'  AGCTAAGGCCTTTCGAATCAAAGGGCCCTGCATAACGGTTGGATCCATTCGCGGCTTATAGGTACCCGTGG  5'
                   1                                        2

                        5'  GGCAATAGCCTAGGAACTTTTAGGCCAATTAA  3'
                        3'  CCGTTATCGGATCCTTGAAAATCCGGTTAATT  5'
                                           3
```

   a. 1 and 2
   b. 1 and 3
   c. 2 and 3

13. Consider the cross AaBb x AaBb. If trait A exhibits complete dominance and trait B exhibits codominance, what phenotypic ratio is expected in the offspring?
   a. 1:1:1:1
   b. 9:3:3:1
   c. 3:6:3:1:2:1
   d. 1:2:1:2:4:2:1:2:1

14. Which of the following would most likely result in a loss-of-function allele?
   a. Lys576Gly
   b. AAG576AAA
   c. AAG576UAA
   d. Deletion in an enhancer

15. Which of the following could result in a dominant negative allele?
   a. Lys576Gly
   b. AAG576AAA
   c. AAG576UAA
   d. Deletion in an enhancer

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16. The alleles that control ABO blood type phenotype demonstrate which of the following inheritance patterns?

    a. complete dominance
    b. recessiveness
    c. codominance
    d. complete dominance, recessiveness, and codominance
    e. none of the above

17. The majority of calico cats are female. However, a few rare males are observed. What is the genotype of a male calico cat?

    a. X_BX_b
    b. X_BY
    c. X_bYY
    d. X_BX_bY

18. A certain disease is caused by homozygosity of the g allele (G is the corresponding wild-type allele). However, the penetrance of the disease is 75%. Two individuals known to be heterozygous have a child. What is the probability that the child exhibits the disease?

    a. 1/4
    b. 3/4
    c. 1/8
    d. 3/16
    e. 9/16

19. An inversion may result from
    a. A 360 degree rotation of a chromosomal region following two double-strand breaks in a chromosome's DNA.
    b. A crossover between DNA sequences present in two position on the same chromosome in opposite orientation.
    c. A crossover between repetitive DNA sequences present in two positions on the same chromosome in the same orientation.
    d. Crossing over between two repetitive sequences on different chromosomes

20. Mutations that result in much less of a protein or a protein with limited function are known as

    a. null mutations
    b. hypomorphic mutations
    c. hypermorphic mutations
    d. conditional mutations
    e. anti-morphic mutations

Part III: Short answer

21. Tag SNPs are used to genotype individuals in large-scale GWAS studies to reduce cost, but these typically do not correspond to SNPs that directly cause the trait or disease of interest. What is the relationship of the Tag SNPs to the causative alleles?

22. The production of purple pigment anthocyanin in flowers is accomplished via a biosynthetic pathway. The pathway is shown:

    Precursor ------→ intermediate --------→ anthocyanin
    Enzyme1 Enzyme2
    A mutation in either one of the enzymes will produce white flowers.
    Enzyme one is coded by gene A (A=functional enzyme and a= non-functional enzyme)
    Enzyme two is coded by gene B (B=functional enzyme and b= non-functional enzyme)

    a. What is accumulated in a plant of genotype aaBB

    b. What will be the color of the flowers on an aaBB plant

    c. What is accumulated in a plant of the genotype Aabb

    d. If an Aabb plant is crossed with a aaBB plant, what would be the color of the flowers in the F1

e. What type of epistatic relationship is this?

23. The following sequence of double stranded DNA encodes a hypothetical protein called GEN in a bacteria. Transcription starts at and includes the C-G base pair in Bold. The underlined T-A base pair indicates the transcription terminator.

```
5′ -TTCCCCTATGGAGGGTCATCTACGAGGCCCCCATCACTAAAGCTTG- 3′
3′ -AAGGGGATACCTCCCAGTAGATGCTCCGGGGGTAGTGATTTCGAAC-5′
```

a. What are the first 9 bases of the transcribed RNA? Be sure to label the 5' and 3' ends of the RNA

b. What are the first 4 amino acids of the subsequent polypeptide?

c. How many amino acids long is the GEN polypeptide?

You isolate a mutant bacterium with the GEN gene sequence below.

```
5′ -TTCCCCTATGGGGGGTCATCTACGAGGCCCCCATCACTAAAGCTTG- 3′
3′ -AAGGGGATACCCCCCAGTAGATGCTCCGGGGGTAGTGATTTCGAAC-5′
```

d. The double underlined G/C base pair is the site of the mutation (a substitution of G/C for A/T in the wild type protein). As before, the bold G/C base pair indicates the start of transcription and the underlined T/A base pair indicates transcription terminator.
What are the first 4 amino acids of the mutant polypeptide?

e. What type of mutation in the coding sequence is this?

24. Polycystic kidney disease is a dominant trait that causes the growth of numerous cysts in the kidneys. The condition eventually leads to kidney failure. A child with polycystic kidney disease is born to a couple, neither of whom shows the disease. What might explain this outcome?

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Part IV: Extended calculations (show your work for partial credit, show pedigrees and genotypes, punnett squares, etc.)

25. You conduct a case-control study comparing the frequency of SNPs among individuals who have developed high cholesterol levels as compared with control individuals of a similar age who show no sign of this condition. Here's the result:

	Cases	Controls
A	23	10
T	13	20

Does a statistically significant association exist between this SNP and the risk of developing high cholesterol, if this is the only SNP that you test? What is the p-value?

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of x^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28

26. Draw a pair of homologous chromosomes as they would be paired during meiosis I if one chromosome had a paracentric inversion. (Be sure to give yourself plenty of space so drawings are clear)

    Let the original gene order on these chromosomes be:
    L, M, N, centromere, O, P, Q, R, S, T

    Let one of the chromosomes contain an inversion that includes P, Q, and R